

Contestant Number: _____

Time: _____

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SQL DATABASE FUNDAMENTALS (345)

STATE – 2020

Multiple Choice & Short Answer Section:

Multiple Choice (50 @ 10 points each) _____ (500 points)

TOTAL POINTS _____ (*500 points*)

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
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No more than sixty (60) minutes testing time

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General Instructions

Your name and/or school name should *not* appear on any work you submit for grading. Write your contestant number in the provided space. Staple all pages in order before you turn in your test.

Assumptions to make when taking this assessment:

Unless specified otherwise, questions are in the context of database systems of the Relational/SQL type database family/genre

MULTIPLE CHOICE

Identify the letter of the choice that best completes the statement or answers the question. Mark A if the statement is true. Mark B if the statement is false.

1. The _____ SQL keyword combination is used to return only unique values.
 - a. Select Distinct
 - b. Select Unique
 - c. Select Distinct Unique
 - d. Select Different

2. The _____ SQL statement may be used with aggregate functions to group the result-set by one or more columns.
 - a. Order By
 - b. Sort By
 - c. Grouping By
 - d. Group By

3. A(n) _____ on a table is typically added to speed up the performance of queries.
 - a. Function
 - b. Trigger
 - c. Column
 - d. Index

4. Which of the below SQL statements will filter to records with FruitName values that are *not* Mango, Apple, or Jackfruit?
 - a. WHERE NOT EXISTS FruitName ('Mango', 'Apple', 'Jackfruit')
 - b. WHERE FruitName DOES NOT CONTAIN ('Mango', 'Apple', 'Jackfruit')
 - c. WHERE FruitName NOT IN ('Mango', 'Apple', 'Jackfruit')
 - d. WHERE FruitName IN ('Mango', 'Apple', 'Jackfruit')

5. What is the purpose of the below SQL?
INSERT INTO Categories (CategoryID, CategoryName)
VALUES(100, 'Electronics');
 - a. Generate the Categories table structure
 - b. Insert two rows into the Categories table
 - c. Create the Categories table and then insert a row into the table
 - d. Insert a row into the Categories table

6. What would be the effect of the below DELETE statement, assuming CITIES is a table containing multiple records.

DELETE FROM CITIES;

- a. The statement removes the last row from the table CITIES;
- b. The statement has no effect on the table CITIES;
- c. The statement removes the first row from the table CITIES;
- d. The statement removes all rows from the table CITIES;

7. What would be the effect of the below DELETE statement?

DELETE FROM Products

WHERE ProductID = 1;

- a. The statement removes any rows from the Products table that have ProductID with value 1.
- b. The statement removes any rows from the Products table that have ProductID starting with 1.
- c. The statement removes all rows from the Products table.
- d. The statement removes no rows from the Products even if there is a row with ProductID = 1.

8. What would be the effect of the below UPDATE statement? Assume Products is a table containing multiple records.

UPDATE Products SET ProductName = 'My Favorite Product';

- a. The statement has no effect.
- b. The statement changes only the first row to have ProductName with value My Favorite Product.
- c. The statement would error, since the UPDATE statement requires a WHERE clause.
- d. The statement changes all rows in the Products table to have ProductName with value My Favorite Product.

9. The _____ statement removes all rows from a table *and* frees the space allocated for the table records.

- a. REVOKE
- b. DELETE
- c. TRUNCATE
- d. ROLLBACK

10. The acronym DCL stands for

- a. Data Control Language
- b. Data Content Language
- c. Database Concept Language
- d. Data Concept Linear

11. Which of the following is an example of DCL?
 - a. INSERT
 - b. CREATE
 - c. COMMIT
 - d. GRANT

12. A Foreign Key constraint *may* be created to link two columns within the *same* table?
 - a. True
 - b. False

13. The Order By keyword sorts records in ascending order by default?
 - a. True
 - b. False

14. 3NF is typically used in the practice of _____.
 - a. Database design
 - b. Data modification
 - c. Data transformation
 - d. Data extraction

15. A _____ is a set of SQL statements with an assigned name stored in the database so that it may be reused.
 - a. View
 - b. Stored Procedure
 - c. Field
 - d. Index

16. A database _____ is procedural code that executes automatically in response to certain events, including DML events.
 - a. Transaction
 - b. Trigger
 - c. Stored Procedure
 - d. Function

17. In the Oracle database management system, which of the following is a procedural language extension to SQL?
 - a. PL/SQL
 - b. Transact-SQL
 - c. C#
 - d. JSON

18. Which of the following database systems is a RDBMS?
- a. Neo4j
 - b. MongoDB
 - c. Couchbase
 - d. Teradata
19. A _____ is a column or a set of columns in a table that uniquely identifies a row.
- a. Fact
 - b. Dimension
 - c. Foreign Key
 - d. Candidate Key
20. A _____ for a table is a unique identifier, for records, that has business meaning and is often derived from application data. International Standard Book Number (ISBN) would be an example.
- a. Surrogate key
 - b. Natural key
 - c. Select Key
 - d. Fact Key
21. Database tables can be created without a Primary Key.
- a. True
 - b. False
22. With respect to a Foreign Key Constraint, the table containing the Foreign Key is the _____ table and the table containing the Candidate Key is the Parent table.
- a. Referenced
 - b. Parent
 - c. Application
 - d. Child
23. Which of the following joins, essentially, combines the results of both left and right outer joins?
- a. All Join
 - b. Inner Join
 - c. Cross Join
 - d. Full Join

24. Which range of values defines SMALLINT?
- a. Integer data from 0 to 255
 - b. Integer data from -32,768 to 32,767
 - c. Integer data from -8,388,608 to 8,388,607
 - d. Integer data from -2,147,483,648 to 2,147,483,647
25. In SQL Server and Postgres, which range of values defines BIGINT?
- a. Integer data from -32,768 to 32,767
 - b. Integer data from -8,388,608 to 8,388,607
 - c. Integer data from -2,147,483,648 to 2,147,483,647
 - d. Integer data from -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
26. In Postgres, which of the below SQL will return the current date and time?
- a. SELECT GETDATE();
 - b. SELECT NOW();
 - c. SELECT TIMESTAMP();
 - d. SELECT GETNOW();
27. The date range for the DATE datatype in Postgres is ____.
- a. January 1, 0001 through December 31, 9999
 - b. January 1, 1000 through December 31, 9999
 - c. 1 BC through 100,000 AD
 - d. 4,713 BC through 5,874,897 AD
28. What would be returned by the below SQL, assuming CityName and CountryName have the same datatype?
- ```
SELECT CityName FROM Cities
UNION
SELECT CountryName FROM Countries;
```
- a. A distinct list of CityName values from Cities that also exist as a CountryName in Countries
  - b. A non-distinct list of CityName values from the Cities table combined with the CountryName values from Countries
  - c. A distinct list of the following: CityName values from the Cities table combined with the CountryName values from Countries
  - d. An error would be returned, as the column name is not the same between the two Select statements

29. Which of the following may be used to perform an inner join between the Customers and Orders tables using their CustID columns?
- a. JOIN Orders ON (Customers.CustID = Orders.CustID)
  - b. INNER JOIN Orders ON (Customers.CustID = Orders.CustID)
  - c. Inner Join Orders ON Customers.CustID = Orders.CustID
  - d. All of the above
30. Which of the below statements adds the column ClientDescription with datatype VARCHAR(10) to the Clients table?
- a. AMMEND TABLE Clients ADD COLUMN ClientDescription VARCHAR(10);
  - b. ALTER TABLE Clients AMMEND ClientDescription VARCHAR(10);
  - c. ALTER TABLE Clients ADD ClientDescription VARCHAR(10);
  - d. MODIFY TABLE Clients ADD COLUMN ClientDescription VARCHAR(10);
31. Which of the following creates a Foreign Key?
- a. CHECK KEY (CategoryID) REFERENCES Categories(CategoryID)
  - b. FOREIGN KEY (CategoryID) REFERENCES Categories(CategoryID)
  - c. FOREIGN KEY (CategoryID) Categories(CategoryID)
  - d. CHECK KEY (CategoryID) Categories(CategoryID)
32. In a RDBMS, a “relation” is physically represented by a \_\_\_\_\_.
- a. Table
  - b. Column
  - c. Foreign Key
  - d. Index
33. In Postgres and MySQL, which SQL statement is used to restrict the number of records returned by a query result set to a specified number of rows?
- a. Top
  - b. Includes
  - c. Limit
  - d. Excludes
34. With respect to Consistency, Availability, Partition Tolerance (CAP), relational databases typically provide \_\_\_\_\_.
- a. Consistency and Availability
  - b. Availability and Partition Tolerance
  - c. Partition Tolerance and Consistency
  - d. Consistency, Availability, and Partition Tolerance

35. A \_\_\_\_\_ is sequence of database operations that satisfies the ACID principles.
- Table
  - Transaction
  - Index
  - Foreign Key
36. Which of the following is not a date/time data type in Postgres?
- SMALLDATETIME
  - TIMESTAMP
  - DATE
  - INTERVAL
37. \_\_\_\_\_ guarantees that a transaction can only bring the database from one valid state to another valid state.
- Atomicity
  - Availability
  - Consistency
  - Partition Tolerance
38. Due to \_\_\_\_\_ once a transaction has completed execution, the updates to the database are persisted in a way that is recoverable upon system failure.
- Durability
  - Isolation
  - Availability
  - Consistency
39. Which of the below SQL statements tests for a column having a null value?
- Is Null
  - Equals Null
  - Contains Null
  - Select Null
40. The BETWEEN operator is inclusive. The begin and end values are included.
- True
  - False
41. A table or column may be given a temporary name within a query by using a(n) \_\_\_\_\_.
- Join
  - Rename
  - Includes
  - Alias

Consider the following SQL Server code for questions 42 – 47. Assume, a Customers table already exists.

```
CREATE TABLE Orders(
 OrderID INT NOT NULL,
 OrderDate DATE NOT NULL,
 OrderTotal DECIMAL(13,4),
 CustomerID INT NOT NULL,
 CONSTRAINT PK_Orders PRIMARY KEY (OrderID),
 FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);
```

```
INSERT INTO Orders Values (1, '01/01/2020', 5.00, 1);
INSERT INTO Orders Values (2, '03/03/2020', 15.00, 1);
INSERT INTO Orders Values (3, '03/03/2020', 10.00, 3);
INSERT INTO Orders Values (4, '05/04/2020', 10.00, 4);
INSERT INTO Orders Values (5, '05/04/2020', 10.00, 1);
```

42. What is the result of the below SQL?

```
SELECT COUNT(*) FROM Orders;
```

- a. 1
- b. 3
- c. 5
- d. 7

43. How many rows are returned by the below SQL?

```
SELECT CustomerID, SUM(OrderTotal) AS OrderTotals
FROM Orders
GROUP BY CustomerID
HAVING SUM(OrderTotal) > 12;
```

- a. 1
- b. 3
- c. 5
- d. 7

44. Which of the below SQL statements will return a single row with the value 3, which is the number of different CustomerID values in the Orders table?

- a. SELECT COUNT(CustomerID) FROM Orders;
- b. SELECT COUNT(DISTINCT CustomerID) FROM Orders;
- c. SELECT DISTINCT CustomerID FROM Orders;
- d. SELECT CustomerID from Orders GROUP BY CustomerID;

45. If additional rows were inserted into the Orders table, which column could accept a NULL value?
- a. OrderID
  - b. OrderDate
  - c. OrderTotal
  - d. CustomerID
46. Which of the below SQL will return the value 5.0000, which is the lowest OrderTotal value?
- a. SELECT SUM(OrderTotal) FROM Orders;
  - b. SELECT MIN(OrderTotal) FROM Orders;
  - c. SELECT FLOOR(OrderTotal) FROM Orders;
  - d. SELECT FIRST(OrderTotal) FROM Orders;
47. Which of the below SQL will return the following results set?

| CustomerID | HighestTotal |
|------------|--------------|
| 1          | 15.0000      |
| 3          | 10.0000      |
| 4          | 10.0000      |

- a. SELECT CustomerID, MAX(OrderTotal) AS HighestTotal FROM Orders GROUP BY CustomerID ORDER BY CustomerID;
  - b. SELECT CustomerID, CEILING(OrderTotal) AS HighestTotal FROM Orders GROUP BY CustomerID ORDER BY CustomerID;
  - c. SELECT CustomerID, HIGHEST(OrderTotal) AS HighestTotal FROM Orders GROUP BY CustomerID ORDER BY CustomerID;
  - d. SELECT CustomerID, TOP(HighestTotal) AS HighestTotal FROM Orders GROUP BY CustomerID ORDER BY CustomerID;
48. In Postgres and SQL Server, what would the below SQL do?
- ```
SELECT * INTO ProductsCopy
FROM Products;
```
- a. Creates a new table called ProductsCopy and then copies the Products table data into that table.
 - b. Copies Products table data into ProductsCopy if ProductsCopy already exists.
 - c. Creates a new table called ProductsCopy and then copies the Product table data into that table and then removes all rows from the Product table.
 - d. Attempts to create a new table called ProductsCopy if it doesn't already exist and then copies the Products table data into that table.

49. As of SQL Server 2017 and earlier, which of the following is not a numeric data type in SQL Server?
- a. TinyInt
 - b. SmallInt
 - c. MediumInt
 - d. Int
50. The date ranges for the DATE datatype in SQL Server, Postgres, and MySQL are identical?
- a. True
 - b. False